

Drava Runtime System

Application Interface

Python / C API Bindings



Pipeline Configuration

Stage definitions, Transport type, Ingress/egress routing, Runtime parameters



Execution Engine

Work-Stealing Thread Pool (XKRT)

I/O Thread

Compute Threads (1..N)

Transport fetch loop

Spawn
Tasks

Execute callback tasks

Dispatch and Batching

Frame accumulation, callback grouping

Atomic Metrics Counters

Throughput, latency, callback timing



Transport Abstraction

Unix Socket
Local IPC

NATS Jetstream
Distributed system